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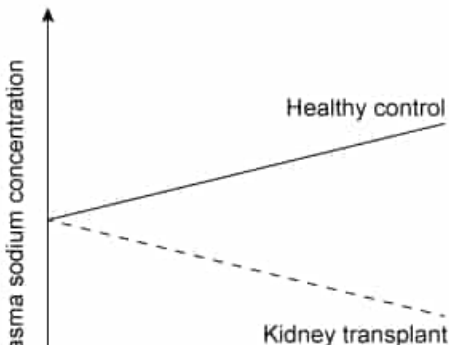
Table 1 Renal and total clearance of superoxide dismutase (SOD)

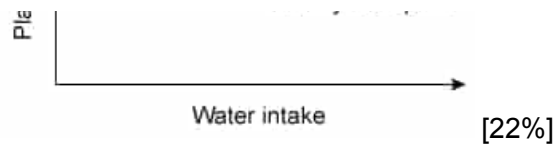
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* significantly greater than renal clearance in 1 mg/kg dose group
** significantly greater than renal clearance in 7.5 mg/kg dose group

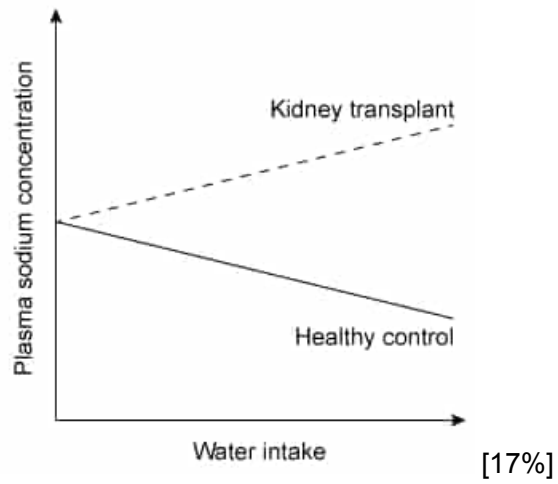
Patients who have undergone kidney transplant surgery often display reduced osmoregulatory capacities compared to healthy controls. Which of the following graphs shows how kidney transplant likely influences the response of plasma sodium concentration to water intake in kidney transplant patients versus healthy controls?

☐ A.

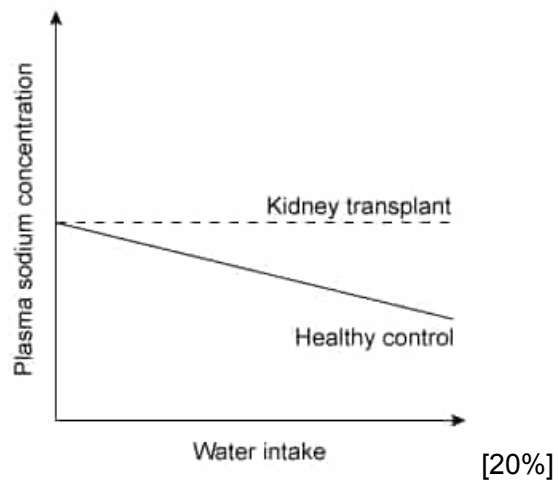




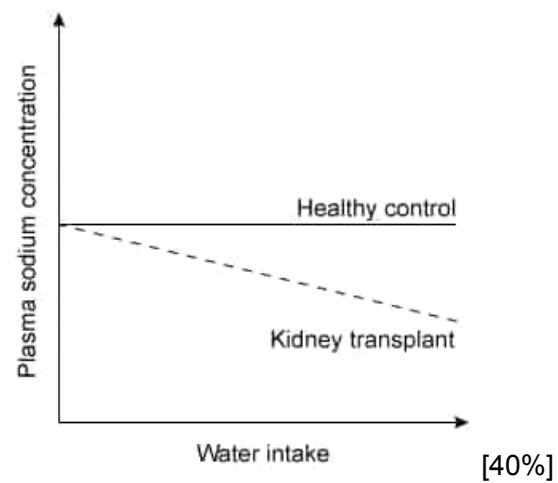
☐ B.



☐ C.



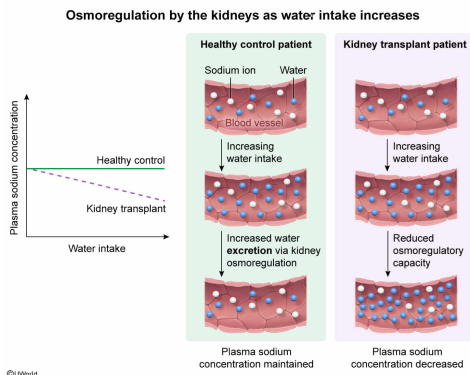
✓ ☒ D.



Correct

39% answered correctly

Explanation:



The **kidney** is a vital organ in **homeostasis**, regulating water and ion content of the blood in a process known as **osmoregulation**. This is accomplished by balancing the intake of specific molecules with the excretion of specific molecules in urine. For example, when large amounts of water are consumed, the **osmolarity** of the blood decreases. To maintain homeostasis in this situation, higher concentrations of solutes (eg, sodium ions) will be reabsorbed, while more water will be excreted by the kidneys, producing a larger volume of dilute urine. In this way, the concentration of water and solutes in the blood stays relatively constant.

The question states that patients with kidney transplants often display **reduced osmoregulatory capacities** compared to healthy control patients. Therefore, as water intake increases, a patient with kidney transplant will likely be unable to adequately perform osmoregulation. In other words, the kidneys are unable to eliminate excess water or reabsorb higher concentrations of solutes. This will lead to a **higher concentration of water in the plasma**, and thus a **lower sodium concentration**. In contrast, healthy patients who can perform osmoregulation will likely display **relatively constant levels of sodium in the plasma**, even as water intake increases. **Choice D** is the only graph to demonstrate this.

(Choice A) Healthy control patients would display relatively constant (*not* increased) plasma sodium concentration as water intake increases.

(Choices B and C) Kidney transplant patients with reduced osmoregulatory capacity would be expected to show decreasing (*not* increasing or constant) plasma sodium concentration as water intake increases.

Educational objective:

The kidney functions in osmoregulation, or the maintenance of consistent electrolyte and fluid concentrations within the body.

Time spent: 0 sec

Subject:
Biology

QID: 404093

System:
Digestion and Excretion

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* significantly greater than renal clearance in 1 mg/kg dose group

** significantly greater than renal clearance in 7.5 mg/kg dose group

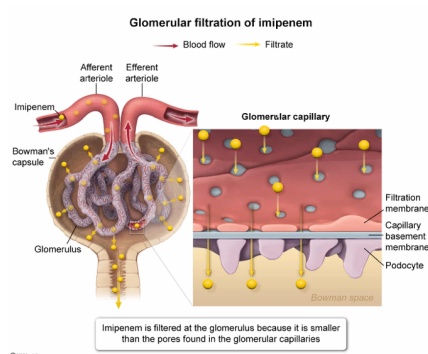
Based on the passage, imipenem is most likely:

- ✓ ☒ A. smaller than pores found in glomerular capillaries. [47%]
☐ B. found only in its unchanged form in the loop of Henle. [8%]
☐ C. too large to be filtered by nephrons. [25%]
☐ D. secreted largely into the distal convoluted tubule. [17%]

Correct

48% answered correctly

Explanation:



The **kidney**, a part of the **urinary system**, filters **blood** and **produces liquid waste** (urine) that is eventually **excreted** from the body. Within the kidney are functional units known as **nephrons**, which perform blood filtration and concentrate urine.

Blood enters the kidneys via the renal **arteries**, which branch into smaller arterioles. These arterioles then branch into the **glomerulus**, the ball-like capillary network in which filtration occurs in each nephron.

The glomerular capillaries contain large pores through which blood passes as it is filtered based on particle size. These capillaries are lined by a basement membrane that allows passage of water, many solutes, and very small proteins into the filtrate. The filtrate is collected by the Bowman's capsule, which surrounds the glomerulus, and is delivered to the proximal convoluted tubule (PCT), the next section of the nephron.

The passage states that the PCT contains peptidases that can metabolize some substances. The antibiotic imipenem is given as an example, a substance metabolized in the PCT and excreted in the urine in a different form from its delivery to the body. Because imipenem is metabolized in the PCT, it must be part of the filtrate that enters the PCT from the glomerulus. Therefore, imipenem likely is filtered at the glomerulus, meaning this compound is **smaller than the pores found in glomerular capillaries**.

(Choice B) The loop of Henle is distal to the PCT. Because imipenem is metabolized in the PCT, it is *not* likely found only in its unchanged form in the loop of Henle.

(Choice C) Because the passage states that imipenem is metabolized in the PCT, it must be small enough (*not* too large) to be filtered by nephrons.

(Choice D) The distal convoluted tubule (DCT) is the site of secretion of some drugs. However, the passage states that imipenem is metabolized in the PCT. Therefore, it is unlikely that this drug is secreted largely into the DCT.

Educational objective:

Substances are filtered through the glomerular capillaries by size.

Time spent:0 sec

Subject:
Biology

QID:404094

System:
Digestion and Excretion

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* significantly greater than renal clearance in 1 mg/kg dose group

** significantly greater than renal clearance in 7.5 mg/kg dose group

Given the data in Table 1, which of the following is LEAST likely to be true?

- ☐ A. The percentage of SOD filtered by the kidneys is higher at doses of 15 mg/kg than at doses of 7.5 mg/kg. [8%]
- ☐ B. Renal clearance of SOD is dependent on dosage. [12%]
- ☐ C. It is difficult to detect SOD in the urine at a dosage of 1 mg/kg. [13%]
- ✔ ☒ D. Total clearance of SOD is dependent on dosage. [65%]

Correct

65% answered correctly

Explanation:

Elimination of superoxide dismutase (SOD) by the body

The **average total clearance** of SOD does not vary markedly among **dosages**; therefore, total clearance of SOD is not likely dependent on **dosage**.

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* significantly greater than renal clearance in 1 mg/kg dose group

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During **urine formation** in the **kidney**, blood is filtered to produce filtrate. The kidneys then reabsorb necessary solutes back into the blood and secrete waste products into the filtrate. **Renal clearance** is a measure of the ability of the kidneys to remove substances from the bloodstream to be excreted.

In the passage, a study was conducted in which the clearance of SOD from the body was measured. At three different drug doses (ie, 1 mg/kg, 7.5 mg/kg, and 15 mg/kg), the average total clearance and average renal clearance (ie, portion of clearance performed by the kidneys) of SOD was calculated.

Table 1 shows that as dosage increases, the average renal clearance increases significantly (**Choice B**). Consequently, a higher percentage of SOD is filtered by the kidneys at doses of 15 mg/kg compared to doses of 7.5 mg/kg (**Choice A**).

Furthermore, the renal clearance of SOD at a dosage of 1 mg/kg is within two standard errors; therefore, it is likely difficult to detect SOD in the urine at this dosage (**Choice C**).

(**Choice D**) The average total clearance of SOD is not indicated as varying significantly between doses. Therefore, total clearance of SOD is *not* dependent on dosage.

Educational objective:

Renal clearance is a measure of the ability of the kidneys to remove substances to be excreted from the bloodstream.

Time spent: 0 sec

Subject:

Biology

QID: 404095

System:

Digestion and Excretion

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* significantly greater than renal clearance in 1 mg/kg dose group
** significantly greater than renal clearance in 7.5 mg/kg dose group

Based on the passage, intracellular transport of drugs eliminated by the liver most likely involves interactions with:

- ☐ A. intermediate filaments. [5%]
- ☐ B. microfilaments. [8%]

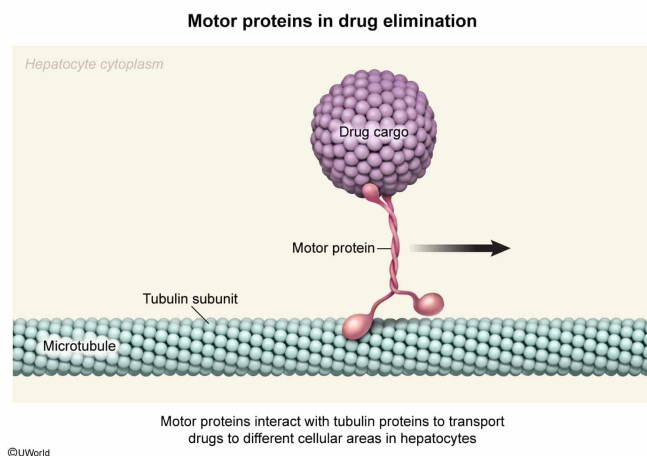
✔ ☒ C. microtubules. [76%]

☐ D. myosin. [9%]

Correct

76% answered correctly

Explanation:



The **cytoskeleton** in **eukaryotic cells** is made of **three families** of protein fibers: **microtubules**, **microfilaments**, and **intermediate filaments**. In addition to supporting the shape and internal structure of the cell, the cytoskeleton provides tracts for the transport of materials within the cell by **motor proteins** (eg, kinesin, dynein). Motor proteins are an example of proteins with nonenzymatic functions.

Microtubules are made of alternating α - and β -tubulin subunits, which assemble into hollow tubes. In addition to their involvement in cell movement as part of the cytoskeleton, microtubules are also involved in cell motility, forming the core of **cilia** and **flagella**, and the formation of the **mitotic spindle** during **cell division**.

The passage states that one route of drug elimination in the body is via the liver. Specialized cells (ie, hepatocytes) endocytose drugs and transport them to different cellular areas using motor proteins. The passage also states that these motor proteins interact with tubulin proteins. Therefore, intracellular transport of drugs eliminated by the liver most likely involves interactions with **microtubules**, because microtubules are made of **tubulin subunits**.

The other choices do not involve tubulin proteins and are therefore either *not* likely involved in intracellular transport of drugs eliminated by the liver or are involved via a mechanism not mentioned in the passage.

(Choice A) Intermediate filaments are made of protein subunits (ie, keratin, lamins, vimentin, and desmin). They are involved in determining the shape of the cell and make up the nuclear lamina; additionally, they help anchor **organelles** to specific areas of the cell.

(Choice B) Microfilaments, thin protein filaments made of actin, are involved in cytokinesis and determining cellular shape. They are also responsible for cellular movement, **muscle contractions**, and movement of vesicles within cells. However, microfilaments are not composed of tubulin and were not otherwise mentioned in the passage.

(Choice D) Myosin is a protein that can function as a motor protein; however, myosin interacts with actin (*not* tubulin) during muscle contraction.

Educational objective:

Cell motility often involves motor proteins that interact with the cytoskeleton.

Time spent:0 sec

Subject:
Biology

QID:404096

System:
Digestion and Excretion

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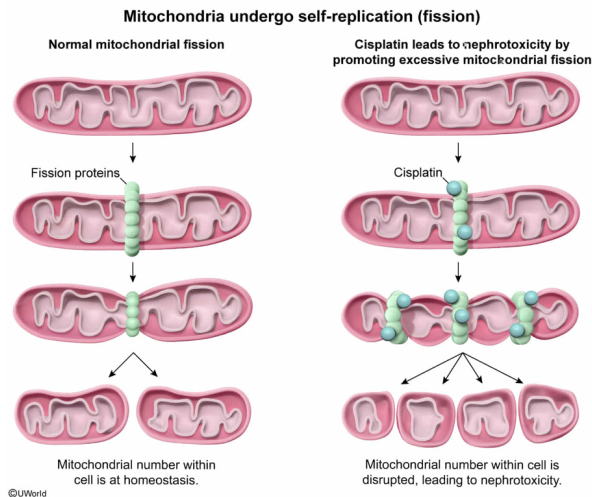
According to the passage, cisplatin most likely directly influences mitochondrial:

- ✔ ☒ A. self-replication. [76%]
- ☐ B. protein synthesis. [3%]
- ☐ C. inhibition of apoptosis. [9%]
- ☐ D. enzymatic reactions. [10%]

Correct

77% answered correctly

Explanation:



Mitochondria are **organelles** with important functions in **aerobic cellular respiration**. A mitochondrion is made of **two layers** formed by two membranes: an outer membrane and a folded inner membrane, the latter of which contains structures essential to the **electron transport chain**.

Between the two membranes is the **intermembrane space**, and within the inner membrane is the **matrix**. **Pumping of protons** to the intermembrane space from the mitochondrial matrix provides the force necessary to generate ATP during **oxidative phosphorylation**. In addition, mitochondria can release enzymes from the electron transport chain to initiate **apoptosis** in a cell.

Mitochondria are somewhat different from other membrane-bound organelles and can be considered semiautonomous. For example, mitochondria **contain their own genes** and have **ribosomes** for **protein synthesis**. Mitochondria are also able to replicate via binary fission in a process that does not directly involve the cell's nucleus.

The passage states that cisplatin is a chemotherapy drug that can lead to nephrotoxicity by promoting excessive fission of mitochondria in renal tubular cells. Therefore, cisplatin most likely directly influences **mitochondrial self-replication**.

(Choices B and D) These choices represent mitochondrial functions that are *not* likely directly prevented by cisplatin.

(Choice C) Mitochondria serve to initiate (*not* inhibit) apoptosis in cells.

Educational objective:

Mitochondria are organelles with roles in cellular metabolism and the initiation of apoptosis. They are semiautonomous, possessing their own genes and self-replicating via binary fission.

Time spent: 0 sec
Subject:
Biology

QID: 404097
System:
Digestion and Excretion